SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1718**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution,

chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and

decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5 Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I SANJAI S(192511018) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SANJAI S

CSA1718 CLOUD COMPUTING

CO3 AT3 RESOLUTION ALGORITHM IMPLEMENTATION

QUESTION 1

RAIN AND WET GROUND KNOWLEDGE BASE
1. IF IT RAINS, THEN THE GROUND BECOMES WET.
2. IT IS RAINING.

GOAL

PROVE THAT THE GROUND IS WET USING THE RESOLUTION ALGORITHM.

TASKS

1. REPRESENT THE STATEMENTS IN PROPOSITIONAL LOGIC.
2. CONVERT ALL STATEMENTS INTO CNF.
NEGATE THE GOAL.
1. APPLY THE RESOLUTION ALGORITHM STEP BY STEP.
2. SHOW THE EMPTY CLAUSE (□) IF THE GOAL IS PROVED.

Absolutely. Since these are Resolution Algorithm numerical/problem-solving questions, I'll structure the answers to match your rubric: problem understanding → method selection → systematic resolution steps → accurate result → interpretation/conclusion.
Question 1 – Rain and Wet Ground
Given Knowledge Base

1. **If it rains, then the ground becomes wet.**
2. **It is raining.**

Goal: Prove that the ground is wet using the Resolution Algorithm.

1. Problem Understanding
Let:

- **R = It is raining.**
- **W = The ground is wet.**

The given statements are:

- **If it rains, then the ground becomes wet.**
- **It is raining.**

We need to prove:
W = The ground is wet.

For resolution, we use proof by contradiction: negate the goal and show that it produces the empty clause (□).

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2. Propositional Logic Representation

Statement 1

"If it rains, then the ground becomes wet."

[
R \rightarrow W
]

Statement 2

"It is raining."

[
R
]

Goal

"The ground is wet."

[
W
]

---

3. Convert Premises into CNF

The implication:

[
R \rightarrow W
]

is equivalent to:

[
\neg R \lor W
]

Therefore, the CNF clauses are:

Clause 1:

[
\neg R \lor W
]

Clause 2:

[
R
]

---

4. Negate the Goal

Original goal:
[
W
]
Negated goal:
[
\neg W
]
Add this as another clause.
Clause 3:
[
\neg W
]
Therefore, the complete clause set is:

Clause	Expression
C1	¬R ∨ W
C2	R
C3	¬W

---

5. Resolution Step-by-Step
Resolution 1
Resolve:
[
C1: \neg R \lor W
]
with:
[
C2: R
]
The complementary literals are:
[
\neg R \quad \text{and} \quad R
]
Therefore:
[
(\neg R \lor W), R
]
gives:

[
W
]
New clause:
C4:
[
W
]

Resolution 2
Now resolve:
[
C4: W
]
with:
[
C3: \neg W
]
The complementary literals are:
[
W \quad \text{and} \quad \neg W
]
Therefore:
[
W,\neg W
]
produces:
[
\boxed{\square}
]
The empty clause is obtained.

---

6. Final Conclusion
Since resolution produced the empty clause (□), the negation of the goal is inconsistent with the knowledge base.
Therefore:
[
\boxed{\text{The ground is wet.}}
]
Goal proved successfully using Resolution.

Question 2 – Student Assignment Submission
Given Knowledge Base

1. **If a student submits the assignment, then the student receives internal marks.**
2. **Rahul submitted the assignment.**

Goal: Prove that Rahul receives internal marks.

---

1. Problem Understanding
Let:

- **S = Rahul submitted the assignment.**
- **M = Rahul receives internal marks.**

The knowledge base states:

- **If Rahul submits the assignment, Rahul receives internal marks.**
- **Rahul submitted the assignment.**

Goal:
[
M
]
We will negate the goal and use resolution to derive the empty clause.

---

2. Propositional Logic Representation
Statement 1
"If Rahul submits the assignment, then Rahul receives internal marks."
[
S \rightarrow M
]
Statement 2
"Rahul submitted the assignment."
[
S
]
Goal
"Rahul receives internal marks."
[
M
]

---

3. CNF Conversion
The implication:

[
S \rightarrow M
]
becomes:
[
\neg S \lor M
]
Therefore:
Clause 1:
[
\neg S \lor M
]
Clause 2:
[
S
]

---

4. Negate the Goal
Goal:
[
M
]
Negated goal:
[
\neg M
]
Add it to the clause set.
Clause 3:
[
\neg M
]
Complete clause set:

Clause	Expression
C1	¬S ∨ M
C2	S
C3	¬M

---

5. Resolution Algorithm
Resolution 1

Resolve:
[
C1: \neg S \lor M
]
with:
[
C2: S
]
Complementary literals:
[
\neg S,\ S
]
Therefore:
[
M
]
New clause:
C4:
[
M
]

Resolution 2
Resolve:
[
C4: M
]
with:
[
C3: \neg M
]
Complementary literals:
[
M,\neg M
]
Therefore:
[
\boxed{\square}
]
The empty clause is obtained.

---

6. Final Conclusion
Since the empty clause (□) has been derived, the negated goal is contradictory with the knowledge base.
Therefore:
[
\boxed{\text{Rahul receives internal marks.}}
]
Goal proved successfully using the Resolution Algorithm.

Question 3 – Library Membership
Given Knowledge Base

1. **If a person is a library member, then the person can borrow books.**
2. **Priya is a library member.**

Goal: Prove that Priya can borrow books.

---

1. Problem Understanding
Let:

- **L = Priya is a library member.**
- **B = Priya can borrow books.**

Given:
[
L \rightarrow B
]
and:
[
L
]
Goal:
[
B
]
We negate the goal and use resolution until the empty clause is obtained.

---

2. Propositional Logic Representation
Statement 1
"If Priya is a library member, then Priya can borrow books."
[
L \rightarrow B
]

Statement 2
"Priya is a library member."
[
L
]
Goal
"Priya can borrow books."
[
B
]

---

3. Convert into CNF
Convert the implication:
[
L \rightarrow B
]
to:
[
\neg L \lor B
]
Therefore:
Clause 1:
[
\neg L \lor B
]
Clause 2:
[
L
]

---

4. Negate the Goal
Original goal:
[
B
]
Negated goal:
[
\neg B
]
Add it to the clause set.

Clause 3:
[
\neg B
]
Complete clause set:

Clause	Expression
C1	¬L ∨ B
C2	L
C3	¬B

---

5. Resolution Step-by-Step
Resolution 1
Resolve:
[
C1: \neg L \lor B
]
with:
[
C2: L
]
Complementary literals:
[
\neg L,\ L
]
Therefore:
[
B
]
New clause:
C4:
[
B
]

Resolution 2
Resolve:
[
C4: B
]

with:
[
C3: \neg B
]
Complementary literals:
[
B,\neg B
]
Therefore:
[
\boxed{\square}
]
The empty clause is obtained.

---

6. Final Conclusion
Since the resolution process produces the empty clause (□), the negated goal is inconsistent with the knowledge base.
Therefore:
[
\boxed{\text{Priya can borrow books.}}
]
Goal proved successfully using Resolution.

Final Answers at a Glance

Question	Knowledge Base	Goal	Empty Clause	Result
Q1	Rain → Wet; Rain	Wet	□	Proved
Q2	Submit → Internal Marks; Rahul submitted	Rahul gets internal marks	□	Proved
Q3	Member → Borrow; Priya is member	Priya can borrow	□	Proved

Resolution Pattern Used in All Three
The general proof strategy is:
[
KB \cup {\neg Goal}
]
Then apply resolution.
If:

$$
\boxed{\square}
$$

is derived, then:

$$
\boxed{KB \models Goal}
$$

So, all three goals are logically entailed by their respective knowledge bases and are successfully proved using the Resolution Algorithm.

Why this should score well against your rubric

- **Problem Understanding (20%) → Variables, premises, and goals are explicitly identified.**
- **Method Selection (20%) → Resolution by contradiction is explicitly selected and justified.**
- **Solution Process (25%) → Propositional representation → CNF → goal negation → resolution → empty clause.**
- **Accuracy (20%) → Each implication is correctly converted as ($P\rightarrow Q \equiv \neg P\lor Q$), and every resolution step is shown.**
- **Interpretation (15%) → The meaning of the empty clause and the final proof are explicitly stated.**